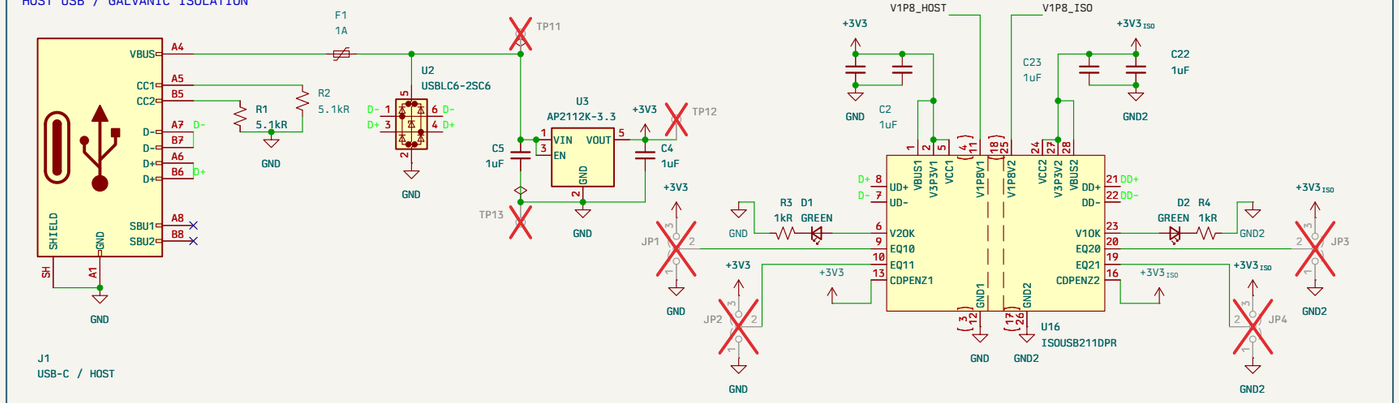
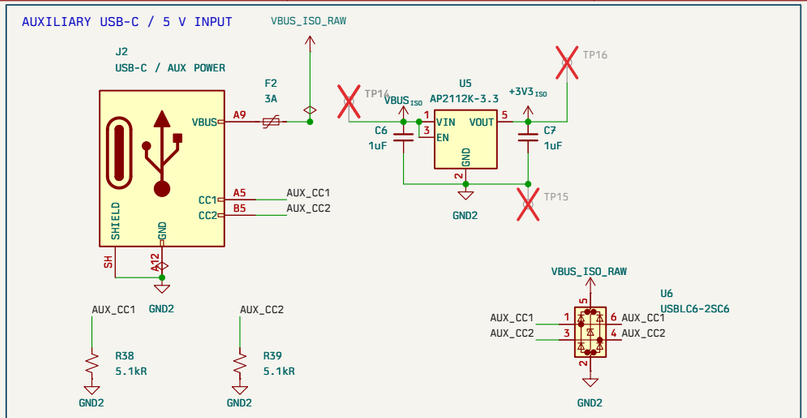
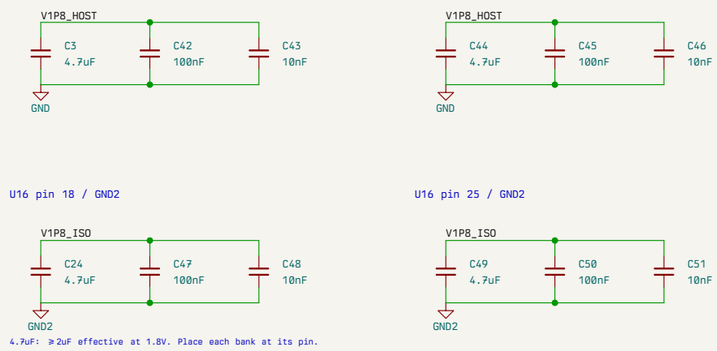
HOST USB / GALVANIC ISOLATION



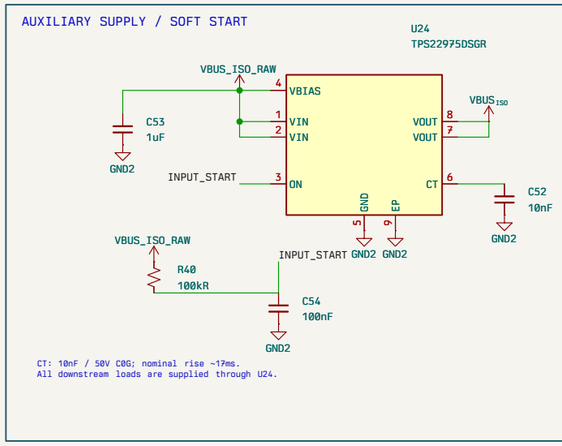
AUXILIARY USB-C / 5 V INPUT



ISOLATOR / LOCAL 1.8 V BYPASS



AUXILIARY SUPPLY / SOFT START



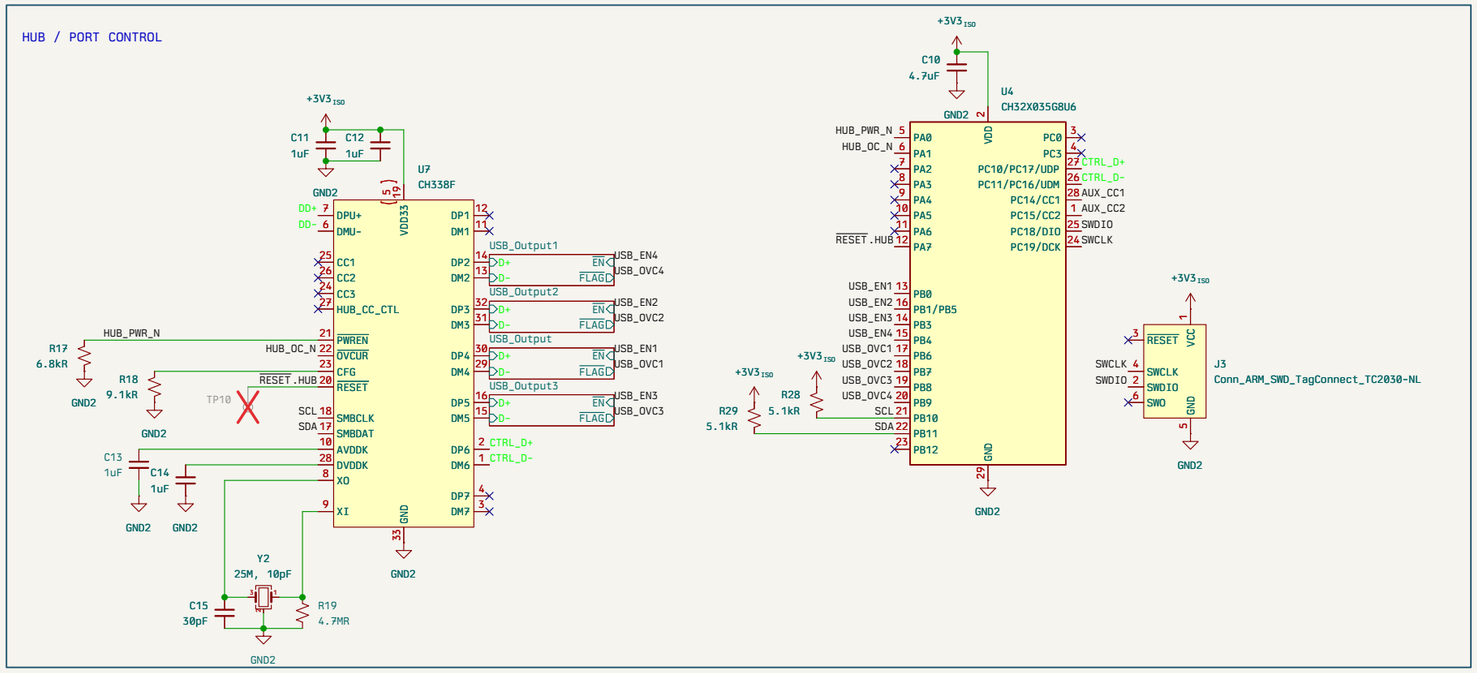
5 V only. Auxiliary supply must preserve isolation.
GND = host side; GND2 = downstream side.

Input CC: AUX_CC1 / AUX_CC2, sensed by U4.
Output CC: local to each repeated port sheet.

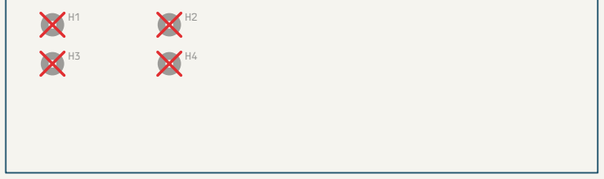
PB10/PB11 use software I2C.
PA0 reads hub power request.
PA1 reports overcurrent (open drain).
Firmware allocates the input-current budget.

WCH DIO / DCK debug; J3 pin 3 unused.
QFN28 PC3 is not external reset.

HUB / PORT CONTROL



MECHANICAL / NOT FITTED



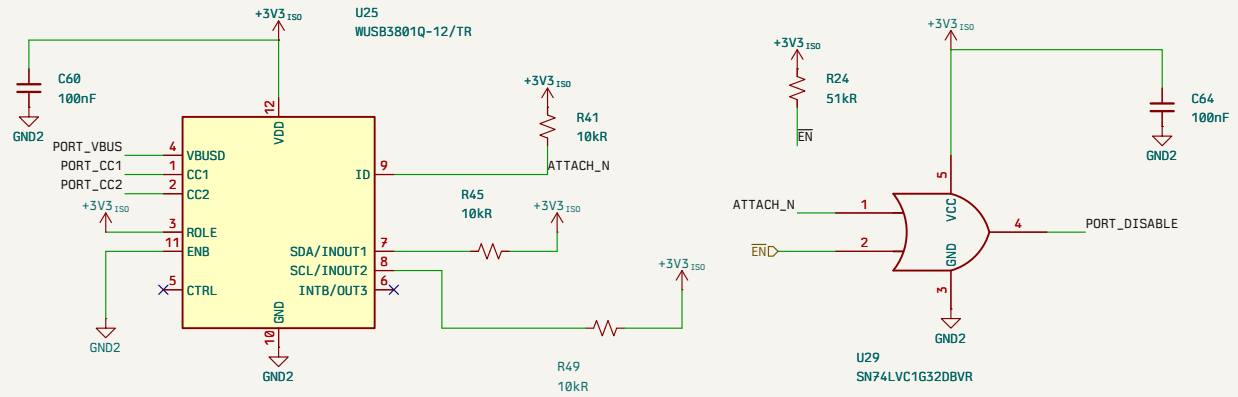
FOR: 5 V only / downstream supply must preserve isolation
American Embedded

Sheet: /
File: fast-computer-condom.kicad_sch

Title: Isolated USB 2.0 Hub / Control and Power

Size: A3	Date:	Rev: B-sch
KiCad E.D.A. 10.0.7		Id: 1/5

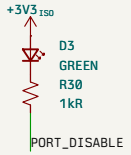
SOURCE ATTACHMENT / ENABLE LOGIC



ROLE=SRC; CTRL floating: pin control.
INOUT2/1 high: default USB current.

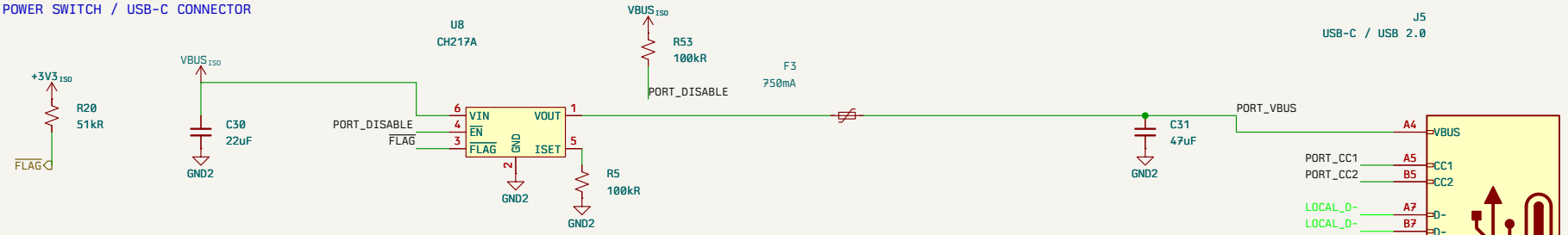
Both conditions must be low:
attached + MCU permission.

STATUS

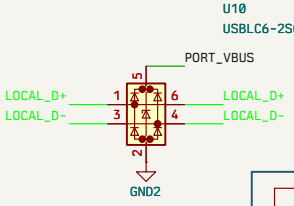


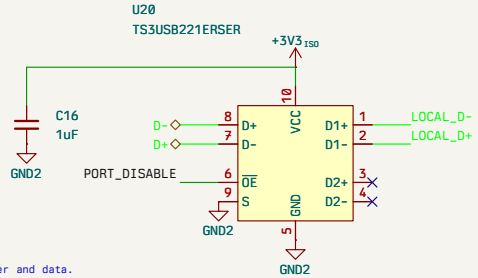
ON

POWER SWITCH / USB-C CONNECTOR



500mA load budget; ~600mA nominal limit.
VIN bypass: ≥ 10uF effective at 5V.
Qualify fuse hold/trip current over temperature.





High on PORT_DISABLE turns off power and data.

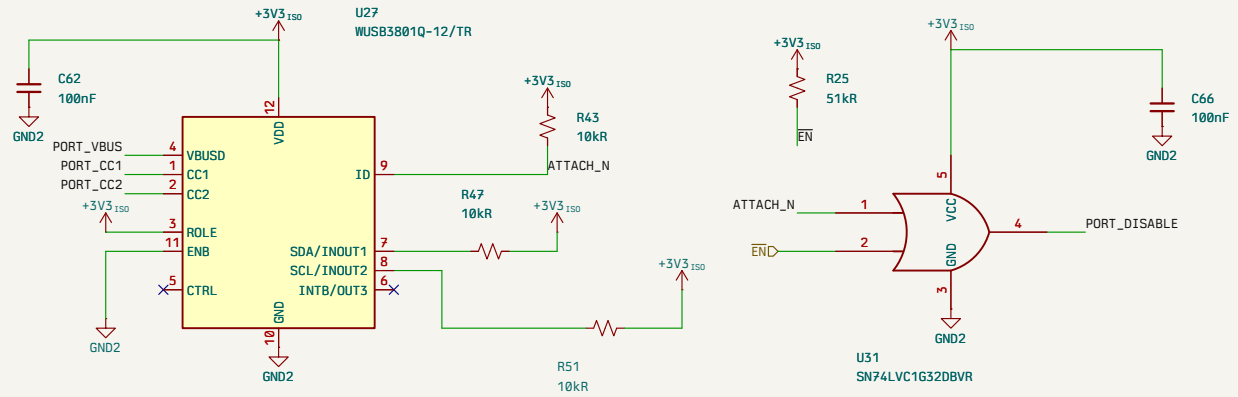
FOR: 5 V / default USB current
American Embedded

Sheet: /USB_Output/
File: USB_Output.kicad_sch

Title: USB-C Output / Attachment-Gated Power and Data

Size: A4	Date:	Rev: B-sch
KiCad E.D.A. 10.0.7		Id: 2/5

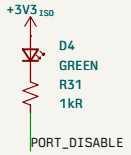
SOURCE ATTACHMENT / ENABLE LOGIC



ROLE=SRC; CTRL floating: pin control.
INOUT2/1 high: default USB current.

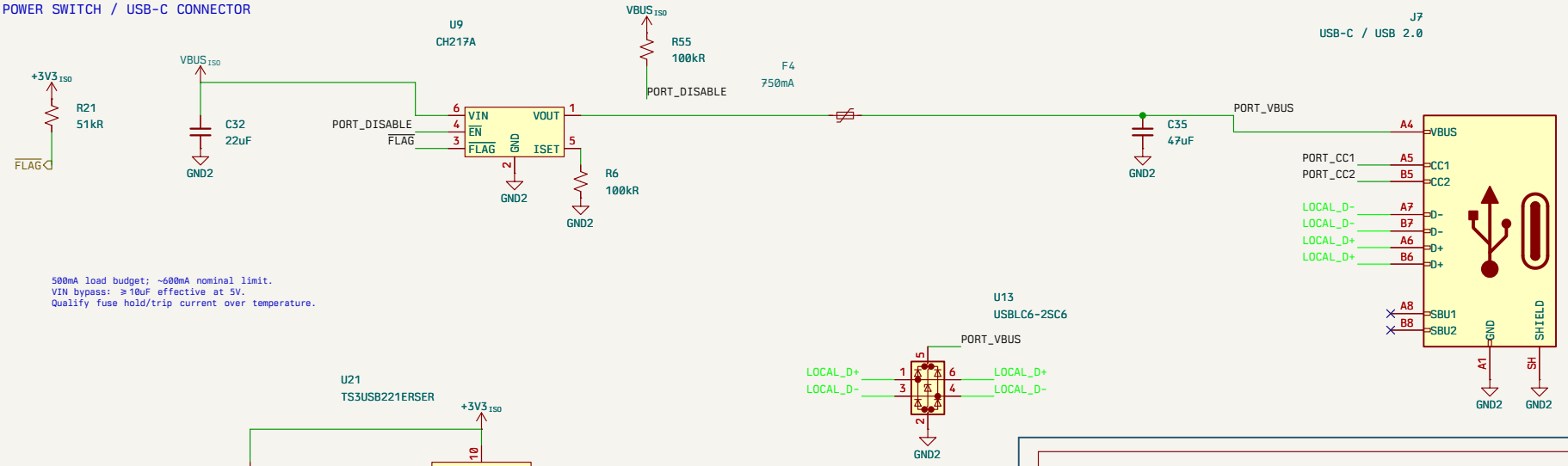
Both conditions must be low:
attached + MCU permission.

STATUS



ON

POWER SWITCH / USB-C CONNECTOR



500mA load budget; ~600mA nominal limit.
VIN bypass: ≥ 10uF effective at 5V.
Qualify fuse hold/trip current over temperature.

High on PORT_DISABLE turns off power and data.

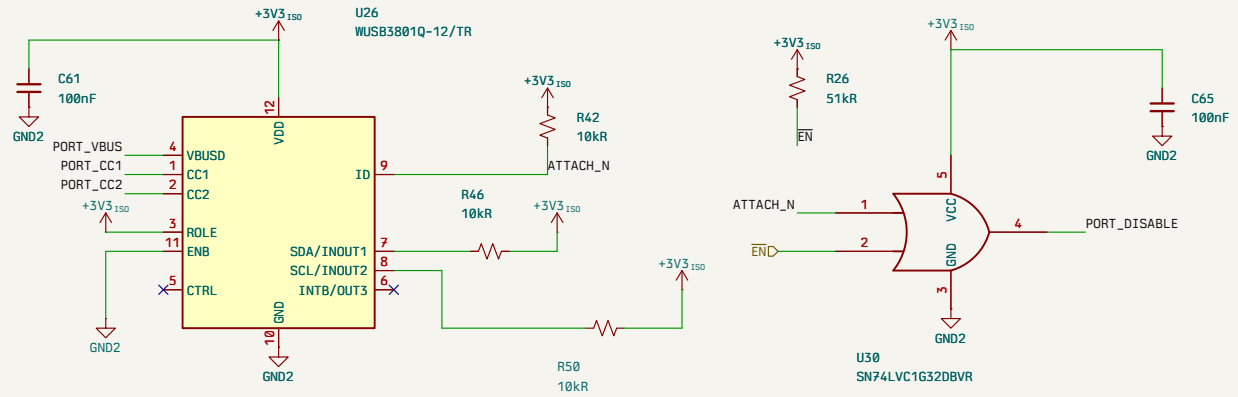
FOR: 5 V / default USB current
American Embedded

Sheet: /USB_Output1/
File: USB_Output.kicad_sch

Title: USB-C Output / Attachment-Gated Power and Data

Size: A4	Date:	Rev: B-sch
KiCad E.D.A. 10.0.7		Id: 3/5

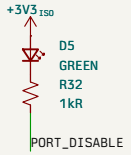
SOURCE ATTACHMENT / ENABLE LOGIC



ROLE=SRC; CTRL floating: pin control.
INOUT2/1 high: default USB current.

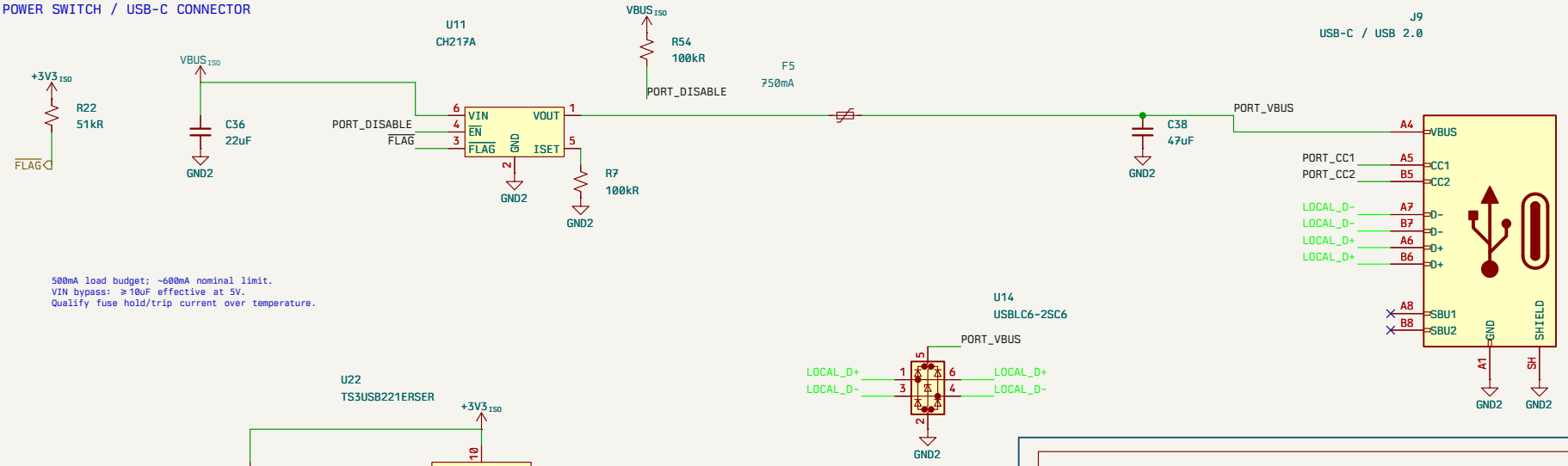
Both conditions must be low:
attached + MCU permission.

STATUS



ON

POWER SWITCH / USB-C CONNECTOR



500mA load budget; ~600mA nominal limit.
VIN bypass: ≥ 10uF effective at 5V.
Qualify fuse hold/trip current over temperature.

High on PORT_DISABLE turns off power and data.

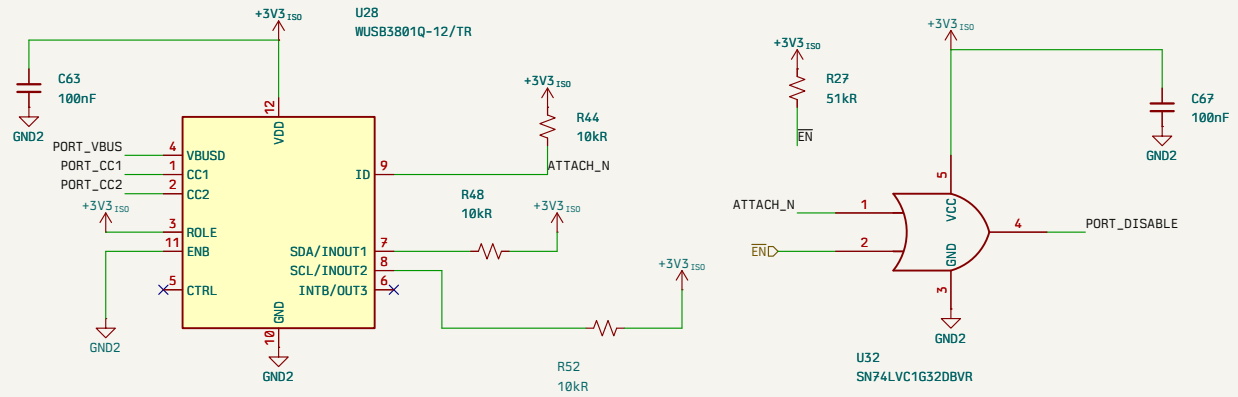
FOR: 5 V / default USB current
American Embedded

Sheet: /USB_Output2/
File: USB_Output.kicad_sch

Title: USB-C Output / Attachment-Gated Power and Data

Size: A4	Date:	Rev: B-sch
KiCad E.D.A. 10.0.7		Id: 4/5

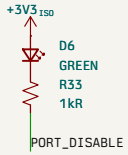
SOURCE ATTACHMENT / ENABLE LOGIC



ROLE=SRC; CTRL floating: pin control.
INOUT2/1 high: default USB current.

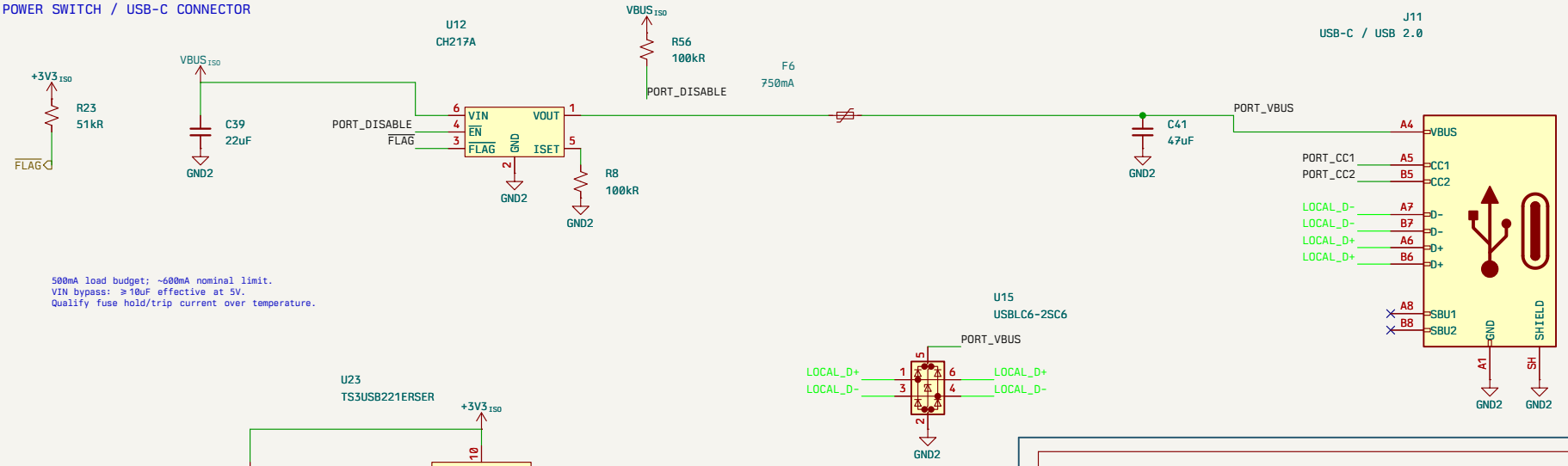
Both conditions must be low:
attached + MCU permission.

STATUS



ON

POWER SWITCH / USB-C CONNECTOR



500mA load budget; ~600mA nominal limit.
VIN bypass: ≥ 10uF effective at 5V.
Qualify fuse hold/trip current over temperature.

High on PORT_DISABLE turns off power and data.

FOR: 5 V / default USB current
American Embedded

Sheet: /USB_Output3/
File: USB_Output.kicad_sch

Title: USB-C Output / Attachment-Gated Power and Data

Size: A4	Date:	Rev: B-sch
KiCad E.D.A. 10.0.7		Id: 5/5